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MDMIE01

# What Did Reality Tell You?

Week 7 • MDMIE01

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THE ARC

# You Are Not Starting Again

Your project should look different today from the idea you had in Week 1.

A

**Week 1 to 4**

Real problem, founder thinking, innovation, the user

B

**Week 5**

Generate and choose

C

**Week 6**

Test the risky assumption

D

**Week 7**

Interpret and decide

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BRING THE EVIDENCE

# Bring the Evidence, Not the Story

**Teams should have**

- Selected idea, riskiest assumption, hypothesis, target users
- Validation method, success signal, failure signal, observations
- Supporting, challenging and unexpected findings, preliminary decision

**Rule**

If you did not run the test, do not invent the result now.

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A SHIFT

# Testing Was Not the Final Step

## Last week

What will we test?

- We designed an experiment card
- We chose a method
- That was preparation

## This week

What did the test actually tell us?

- Running an experiment is activity
- Changing a decision is learning
- That is the real work

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ISA PRESENTATION

# Presentation 1 Is a Decision Story

*By Presentation 1, be ready to say what you selected, tested, learned, and changed.*

| Type | Deadline | Format | Marks |
| --- | --- | --- | --- |
| **Group, every member participates** | **End-August target window** | **5 slides, feedback-focused** | **Indicative 10** |

**Note:** *Nothing changed is acceptable only if the evidence genuinely supported the direction.*

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MODULE 3 ENDS TODAY

# Where We Are

## Week 5 and 6

Brainstorming, SCAMPER, mind mapping

- Idea evaluation
- Assumptions and 5 Whys
- Hypotheses and validation design

## Week 7, today

Interpret evidence and revise decisions

- Challenge problem statements
- Close Module 3
- Next: Business Case Presentation

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FOUR HOURS

# Today's Four-Hour Flow

Five formal activities only.

A

**1. Read**

Read your evidence properly

B

**2. Find**

Find patterns, not convenient quotes

C

**3. Decide**

Decide what changes

D

**4. Learn**

When to improve, pivot or stop

E

**5. Stress-test**

A real SIH 2026 problem statement

F

**6. Rehearse**

Final Presentation 1 rehearsal

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PART 1

# Evidence Must Change Something

From observations to decisions.

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PREVENT OVERCLAIMING

# A Test Result Is Not a Verdict

*Five users are not the entire market.*

One successful test does not prove product-market fit, commercial viability, scalability or long-term adoption. It tells you something narrower.

**Say this**

This test supports the usability hypothesis, not our startup is validated.

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SEPARATE THE STEPS

# Observation Before Interpretation

*Record what happened before explaining why.*

Observation: three users opened WhatsApp after seeing our prototype. Interpretation: they may still prefer their existing channel.

**Bad jump**

Students hate apps, is not what you actually observed.

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THREE BUCKETS

# Support, Challenge or Unclear

User completes a task in 70 seconds supports usability. Cannot find the action challenges it.

A

**Supports**

Evidence behaves roughly as expected

B

**Challenges**

Evidence conflicts with an important assumption

C

**Unclear**

The result does not tell you enough

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IN-CLASS ACTIVITY

# Evidence Autopsy

20 min Duration

Teams Group size

1. Take your Week 6 test results and create three columns
2. SUPPORTS, CHALLENGES, UNCLEAR. Place every observation into one column
3. What did you directly observe, and what did users only say
4. What surprised you, and which result matters most

**Expected output**

One evidence table plus one most important finding

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EVIDENCE QUALITY

# Nice Idea Is Not a Result

*A compliment is not evidence of behaviour.*

Weak: they liked it, nobody complained, my friend said he would download it. Stronger: user completed the task, returned, or booked.

**Watch for**

Anything that feels positive but describes no actual action.

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REPETITION MATTERS

# Pattern or Anecdote?

## Anecdote

One person does something

- Example: 1 of 5 cannot find the button
- Investigate further
- Not yet a pattern

## Pattern

The same behaviour appears repeatedly

- Example: 4 of 5 cannot find the button
- A strong usability signal
- Still not population-level proof

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WORKED EXAMPLE

# Five Students, Five Results

*Do not change the rule after seeing the result.*

Hypothesis: at least 4 of 5 find a verified internship in under two minutes. Results: 1:10, 1:35, could not complete, 1:50, 3:15. Only 3 completed under two minutes.

**Answer**

No, the success rule did not pass.

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CONFIRMATION BIAS

# But They Looked Interested

*Do not negotiate with your own experiment after it finishes.*

The team expected success, the result failed, so they start finding other reasons to call it successful: they smiled, they asked questions, one user was confused anyway.

**Watch for**

Reframing failure as success after the fact.

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MIXED RESULTS ARE NORMAL

# Contradictory Evidence Is Normal

*A failed component does not always mean the entire problem is false.*

Users understand the prototype, but users do not want another platform. Both can be true: the value is useful, but the delivery channel is wrong.

**Student takeaway**

Look for which part failed, not whether everything failed.

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FOUR DECISIONS

# KEEP / CHANGE / DROP / TEST AGAIN

Keep does not mean finished.

**A**

**KEEP**

Evidence supports the current direction

**B**

**CHANGE**

The problem may be real, but something must change

**C**

**DROP**

A major assumption appears false

**D**

**TEST AGAIN**

Evidence is too unclear to decide

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EVIDENCE PLUS CONSEQUENCE

# How Strong Is Your Decision?

*How important was the assumption, and how clear was the evidence?*

Was behaviour observed? Did the result repeat? What happens if we are wrong?

**Example**

A wrong button colour is low consequence. A wrong payment warning is high consequence.

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IN-CLASS ACTIVITY

# Decision Board

20 min Duration

Teams Group size

1. Write your original assumption
2. Write your strongest supporting and challenging evidence
3. Choose one decision: KEEP, CHANGE, DROP or TEST AGAIN
4. Write exactly two sentences explaining why

**Expected output**

One evidence-backed team decision

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NOT A BUZZWORD

# Improvement Is Not Always a Pivot

## Improve

Same user, same core problem

- Same basic direction
- Better execution
- Example: better search filters

## Pivot

One important part changes

- Evidence changed what you believe is true
- Example: WhatsApp, since users refuse another app
- A real direction change, not just a tweak

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FIVE CATEGORIES

# A Direction Can Change

What changed, and what evidence forced that change?

**A**

**User**

A different target group

**B**

**Problem**

A different pain is more important

**C**

**Solution**

A different way of solving it

**D**

**Channel**

Reach or deliver differently

**E**

**Scope**

A narrower part solved first

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CASE STUDY

# Swiggy Toing and Snacc

*Experiments are allowed to lose*

## What happened

Swiggy describes Toing and Snacc as experiments using different business models inside its Platform Innovations segment, a deliberate sandbox for testing new consumer propositions.

## Why it matters

Large companies also run experiments whose outcomes are not predetermined. Do not teach that Swiggy knew these would work.

**Takeaway**

*Swiggy deliberately tested different propositions, not a single guaranteed plan.*

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CASE STUDY

# Snacc

*Stopping can be a good decision*

## What happened

Swiggy said it launched, tested and shut down Snacc within FY26. The micro-kitchen model required high demand density, and the company was not convinced that the steady-state category size justified the additional operating complexity.

## Why it matters

A test that tells you not to continue can save more value than a test that confirms what you wanted to hear.

**Takeaway**

*Was shutting Snacc a failure of experimentation, or evidence that experimentation worked?*

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CASE STUDY

# Food on Train

*Not every experiment ends the same way*

## What happened

A different Swiggy service, Food on Train, expanded to more than 180 cities and reported 3x year-on-year order growth for April to June 2026.

## Why it matters

Snacc stopped. Toing remained under experimentation in Swiggy's FY26 disclosure, while Food on Train later expanded. Different propositions can produce different decisions.

**Takeaway**

*Different evidence and economics can lead to different decisions.*

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GROUP ACTIVITY

# Pivot or Improve?

12 minutes, teams plus class vote. Classify six situations as improve, change, drop or test again.

### Improve cases

Users complete the task but need one label changed

### Change or drop cases

Users refuse another app, or the assumed customer is wrong

### Test again cases

Results are mixed after only three poor-quality tests

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RECAP

# Part 1 Recap

## You now know

- Observation before interpretation
- Pattern versus anecdote

## You practised

- Support, challenge or unclear
- Decision rules matter

## Remember

- A negative result can be useful
- Change should follow evidence

Ready for Part 2: real problems, and how to read them properly.

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PART 1 IS DONE

# Break

Now we use the same thinking on a real national problem.

**Short break**

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PART 2

# Real Problems Are Messy

Reading and challenging Smart India Hackathon problem statements.

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A PROBLEM BANK

# SIH Is a Problem Bank

*Even if you never enter SIH, the problem statements are excellent practice material.*

The SIH 2026 spreadsheet snapshot from 21 August 2026 contains 226 problem statements in total, made up of 172 software and 54 hardware statements.

**Source**

SIH is a platform where student teams solve problems from ministries, departments and industries.

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TEN QUESTIONS

# How to Read a Real Problem Statement

A 1,000-word problem statement can still contain an untested assumption.

**A**

### Who and what

Who has the problem, what happens today, what is painful

**B**

### Evidence given

What evidence or context is given

**C**

### What is asked

What solution is requested, what assumptions are hidden

**D**

### What is missing

Constraints, missing information, what could be tested cheaply

**E**

### Success

What would success actually mean

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CASE STUDY

# SIH26018

*Intelligent land record digitisation*

## What happened

Ministry of Rural Development. An AI platform to extract and validate information from scanned records, handwriting, maps and legacy PDFs in Indian languages.

## Why it matters

It also asks for integration with land-record and GIS systems. Submission deadline as of the 21 August snapshot: 20 September 2026, subject to change.

**Takeaway**

*This is a real, current, and genuinely hard government problem.*

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A REAL PROBLEM

# This Is a Real Government Problem

*A problem can remain difficult even when a lot of digitisation has already happened.*

The current DoLR public page reports: 95.09% of Records of Rights computerised, over 68% cadastral-map digitisation, and over 93% registration computerisation, based on implementation data as of 31 December 2023.

**Teaching point**

Digitisation alone does not solve consistency, interoperability and integration problems.

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A HUGE BRIEF

# The Requested Solution Is Huge

Should a student team try to perfect all of this before testing anything? No.

**A**

**Recognition**

Multilingual OCR, handwriting recognition, field extraction

**B**

**Validation**

Classification and cross-database checks

**C**

**Trust layer**

Confidence scores, human verification, learning over time

**D**

**Systems**

GIS, LRMS, DILRMP, dashboards, APIs, audit trails

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APPLY WEEK 6 THINKING

# Requested Is Not Proven

*Technology named in a problem statement is not evidence that every technical choice will work.*

Which part creates the most value? What is technically risky? What data exists? What accuracy is acceptable? Who verifies errors?

**Ask**

How poor are the scans, and what happens when confidence is low?

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MORE THAN ONE USER

# Who Is the User?

Government is not a customer persona.

**A**

**Staff**

Land-record staff and verification officers

**B**

**Administrators**

District, state and system administrators

**C**

**Citizens**

Citizens seeking accurate records

**D**

**Departments**

Departments consuming land-record data

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RISKY ASSUMPTIONS

# What Could Kill This Solution?

Teaching hypotheses, not claims that SIH has confirmed these failures.

**A**

### OCR accuracy

Can it read degraded handwriting accurately enough

**B**

### Ground truth

Does validation data actually exist

**C**

### Confidence

Do scores correlate with actual errors

**D**

### Integration

Do legacy systems expose usable interfaces

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SCOPE DISCIPLINE

# What Would a Small Test Look Like?

*This is a classroom validation design, not an official SIH requirement.*

Take 20 sample records, 2 languages, 3 document-quality levels, 5 important fields. Test: can it extract the fields, which fields fail, does confidence identify uncertain extraction?

**Rule**

Do not build the nation-wide platform first.

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IN-CLASS ACTIVITY

# SIH Problem Statement Red Team

25 min Duration

Teams Group size

1. Choose SIH26018 or another SIH 2026 problem statement
2. Write the specific user and the real problem in one sentence
3. Write five hidden assumptions and three important constraints
4. Name one assumption that could kill the idea, and one smallest test

**Expected output**

One problem-statement red-team sheet, plus one claim you would not make yet

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EIGHT QUESTIONS

# Do Not Choose Just Because It Is Cool

Difficulty is not the same thing as winning potential.

**A**

**Domain**

Can we understand it, and access relevant data

**B**

**Access**

Can we reach a real user or domain expert

**C**

**Scope**

Is there a reasonable prototype scope

**D**

**Measure**

Is there a clear success measure

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EVIDENCE LITERACY

# Read the Actual Brief

*Evidence literacy also applies to your sources.*

A third-party copy of the SIH26018 listing mislabels its theme as MedTech, though the actual problem is about land records.

**Teaching takeaway**

Metadata can be wrong, stale or mismatched. Always read the full problem statement and verify important fields from the original source.

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WRITE IT PROPERLY

# Bring This Discipline to Your Idea

*A strong statement makes the user, problem, workaround, consequence and evidence clear.*

Weak: students need an AI productivity app. Stronger: second-year students miss relevant internship deadlines because verified updates are scattered across channels.

**Test it**

Would a stranger picture the exact user and situation?

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STILL FIVE SLIDES

# Your Five Slides Remain Fixed

Still five slides.

A

**Slide 1**

Problem, context and target user

B

**Slide 2**

Customer evidence and direct observation

C

**Slide 3**

Journey, pain point and HMW

D

**Slide 4**

Selected direction, innovation type, alternatives and prototype

E

**Slide 5**

Why this direction, assumption, test, result, decision, what changed

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WHAT WE WANT TO HEAR

# Because of the Evidence, We Changed

*Weak: everyone liked our idea, so we continued.*

Strong: we kept the problem but changed the delivery channel. Our test supported usability but challenged repeated-use assumptions. We narrowed to one user group.

**Also strong**

The result was unclear, so we are testing again.

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GROUP ACTIVITY

# Presentation 1 Red-Team Rehearsal

20 minutes, pair two teams. Team A gets 3 minutes to explain its five-slide story.

### Team A explains

Its five-slide story in 3 minutes

### Team B questions

2 minutes: what changed, what is uncertain, what claim lacks evidence

### Then switch roles

Attack the reasoning, not the team or their confidence

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BE READY

# Questions You Should Be Ready For

Prepare an honest answer, not a confident one.

A

**The problem**

Why is it worth solving, and why this user

B

**The evidence**

What evidence do you have, generated by real people

C

**The test**

What was your riskiest assumption, how did you test it

D

**The result**

What happened, what changed, where did AI help

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ASSIGNMENT

# Final Evidence-to-Presentation Pack

*Make the final five-slide Presentation 1 evidence-based.*

| Type | Deadline | Format | Marks |
|---|---|---|---|
| **Permanent team** | **Before Presentation 1, batch-specific deadline on Teams** | **Maximum 2 handwritten pages** | **N/A unless notified** |

**Note:** *Scan as one PDF, Microsoft Teams only. File name: Team Name - MDMIE01 Week 07 Assignment.*

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TWO PAGES

# What the Two Pages Must Contain

### Page 1: what reality told us

- Idea, assumption, hypothesis, test, user count, supporting and challenging evidence
- Unexpected finding, KEEP/CHANGE/DROP/TEST AGAIN, reason, what changed
- Page 2: one line per slide, three expected questions and answers, one claim removed, SIH Scout Box

### File name

Team Name - MDMIE01 Week 07 Assignment.pdf

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RECAP

# Module 3 Is Complete

## Concept

- Problem, then evidence, then alternatives
- Choice, assumption, test, result, decision

## Example

- Swiggy tested and made different calls
- SIH26018 taught scope discipline

## You did

- Practised evidence autopsy and decisions
- Rehearsed and red-teamed Presentation 1

Defend your decision-making, not your original idea.